

DIPESH BIST

Full Stack Developer with professional experience building and maintaining production IoT platforms, AI-powered applications, enterprise web systems, and Shopify e-commerce solutions. Skilled in backend development, frontend engineering, database management, cloud deployment, Docker containerization, and CI/CD automation. Currently working at Nepal Digital Systems Pvt. Ltd., delivering scalable and secure solutions using NestJS, React, TypeScript, PostgreSQL, Docker, and modern software development practices.

✉ dipeshbist150@gmail.com

☎ 9768495902

📍 Swayambhu, Kathmandu

🌐 feed

🐙 Dipeshbist

EXPERIENCE

Nepal Digital Systems Pvt. Ltd.

Intern to Full Stack Developer

Jan 2025 - Present

Garud IoT Platform (Production System)

- Designed, developed, and deployed a production IoT telemetry platform for monitoring devices and real-time sensor data.
- Built scalable backend services using NestJS, TypeScript, Prisma ORM, and PostgreSQL.
- Implemented secure authentication and authorization using JWT and role-based access control.
- Integrated ThingsBoard for device management and telemetry processing through MQTT/MQTTS communication.
- Configured mTLS security certificates including Root CA, Device Certificates, and Private Keys for secure device connectivity.
- Developed responsive dashboards using React, Vite, TailwindCSS, Recharts, and React Leaflet.
- Implemented file upload and email notification systems using Cloudinary and Nodemailer.
- Dockerized services and managed deployments using Docker Compose, Nginx, SSL certificates, and VPS hosting.
- Automated deployment workflows using GitHub Actions CI/CD pipelines.

Nepal Digital Systems Website

- Developed and deployed the official company website from development to production.
- Built frontend and backend modules using React, Express, and PostgreSQL.
- Configured Docker containers, Nginx reverse proxy, SSL certificates, and VPS hosting.
- Implemented CI/CD pipelines for automated deployment of frontend, backend, and admin services.

SmartCap Shopify Store

- Developed and customized a Shopify storefront using Shopify Liquid and Dawn theme architecture.
- Built responsive product pages, navigation menus, custom sections, and storefront components.
- Configured products, collections, theme settings, and store functionality to meet business requirements.
- Worked on Shopify theme customization, responsive design, and e-commerce user experience improvements.

EDUCATION

International School of Science and Management (ISMT)

BSc (Hons) Computer Systems Engineering

Sep 2023 - Present

Partnered with University of Sunderland, UK

Radiant Secondary School

+2 Science Stream - 3.4

June 2021 - June 2023

PROJECT

Academic & Personal Projects

Diabetes Prediction + RAG AI System

- **Technologies:** NestJS, React, Python, ONNX, Groq API, PostgreSQL, Prisma
- Developed an AI-powered healthcare platform combining machine learning prediction and Retrieval-Augmented Generation (RAG).
- Built a Random Forest model achieving 96.6% prediction accuracy for diabetes risk assessment.
- Deployed the machine learning model using ONNX Runtime directly within NestJS for efficient production inference.
- Implemented a medical knowledge assistant capable of answering questions from healthcare documents using RAG architecture.
- Integrated Groq LLMs, local embeddings, and MCP tools to enhance AI capabilities.

SecureAuth System

- **Technologies:** Express.js, Node.js, PostgreSQL, JWT, Google reCAPTCHA
- Developed a secure authentication and authorization system supporting registration, login, password recovery, and session management.
- Implemented JWT access and refresh token architecture with role-based access control.
- Integrated Google reCAPTCHA v3 to protect against automated attacks and abuse.

Taskboard Kanban Task Manager:

- **Technologies:** React, Express.js, PostgreSQL
- Developed a full-stack Kanban task management application supporting drag-and-drop workflow management.
- Built authentication, task assignment, and complete CRUD functionality.
- Designed RESTful APIs and database structures for task tracking and collaboration.

AI Client Inquiry System:

- **Technologies:** Express.js, React, PostgreSQL
- Built an AI-powered inquiry management platform for intelligent customer query handling and routing.
- Developed frontend and backend modules supporting inquiry tracking and management workflows.

FitLife Android Fitness App:

- **Technologies:** Kotlin, Jetpack Compose, SQLite
- Developed a native Android application enabling workout tracking, goal management, and activity monitoring.
- Designed SQLite database architecture and implemented application business logic.

SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python, Kotlin
- **Frontend Development:** React, Vite, HTML, CSS, TailwindCSS, Bootstrap, shadcn/ui, Recharts, React Leaflet
- **Backend Development:** Node.js, NestJS, Express.js
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Neon
- **Authentication & Security:** JWT, Passport.js, bcrypt, RBAC, Google reCAPTCHA v3
- **AI & Machine Learning:** scikit-learn, ONNX, RAG, Groq API, Xenova Embeddings, MCP Server
- **DevOps & Cloud:** Docker, Docker Compose, Nginx, VPS, GitHub Actions, CI/CD, SSL
- **IoT Technologies:** ThingsBoard, MQTT/MQTTS, mTLS Certificates
- **Shopify Development:** Shopify Liquid, Theme Customization, Product Configuration, Store Development
- **Tools & Services:** Prisma, Cloudinary, Nodemailer, Git, GitHub

HOBBIES/INTERESTS

- Learning emerging technologies
- IoT & Industrial Automation
- Cloud & Backend Architecture

LANGUAGE

Nepali Advanced

English Advanced